

Techno-judges on the spot: explorations of the impact of AI advancements in Brazil's Justice System

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This paper aims to analyse the effects of the implementation of AI decision-making tools in Brazil's Justice System, focusing on judge's work in two main courts in São Paulo, the State Court of São Paulo (TJSP) and the Federal Court of the Third Region (TRF3R). Considering the overall rise in AI adoption in public services, regardless of the area, stands the fact that there's an enormous heterogeneity in all categories of AI use and application, as well as heterogeneities in the implementation process, the choice and development of such algorithmic models [17]. These models and tools are usually developed and driven by the bureaucracy's limited capabilities or through contracts of licenses, LLMs and other cloud-based services provided by Big Tech companies. AI tools may strengthen administrative capabilities and promote social inclusion in public policies [1][7], but there's usually ethical and structural challenges, as well as multiple risks, such as algorithmic biases being made and reproduced through data and the probability of bureaucrats' discernment being drastically reduced, which could entail in amplifying inequalities and compromising general public's access to public services [9][4][16]. Regarding Justice Systems, judges tend to use AI tools to accelerate the decision-making process, with the goal of reaching higher productivity metrics [14][10], especially in Brazil's case where the whole Justice System is overloaded in the number of pending cases, accounting a backlog of over 80 million. Thus, AI appears to these institutions and courts as a miraculous tool that will fix most, if not all, of their problems regarding speed, efficiency and productivity in the decision-making process. However, most of the literature regarding this theme focuses on the Global North. Even though there's some literature in Brazil that looks at the effects of AI in the public sector [17][5], in regards to the Judiciary, the reference works do not tackle sociological investigations on this matter, focusing solely on transparency and governance in such contexts [12]. As such, the analysis of Brazil's case fills an important gap in this discussion, by taking a look at different organizational contexts, which show a broader picture of public institutions' capabilities and fragilities, especially regarding Justice Systems, since these differences play a main role in affecting the bureaucracies' work [15]. In addition to this, this paper tackles another fundamental gap in the literature, of which there's very few works taking a qualitative approach in social sciences to evaluating the impacts and perceptions of the users of these tools, specifically works that delve in bureaucrats, public servants and judges' day to day work [16][10]. We aim to contribute to this debate by evaluating the degree and the type of transformations perceived in judges' position and function, as well as in jurisdictional work before and after the implementation of such tools in their daily decisions' routines. As such, the main question this paper wants to answer is: have decision-making support tools implemented in courts changed judge's overall work and discretionary role? If so, how? For our research, we chose a period of five years, from 2020 to 2025, which proves to be a very important and transitional period for Brazilian Courts' digital transformation, especially considering the outcomes from the COVID-19 period, where all public services had to be digitalized to keep them available to citizens. Countless tools such as administrative automations, electronic complaint systems and over 140 AI projects [10], currently finished or in late development stages, show a progressive leap in the Judiciary's technological prowess. These projects have, although, matured extremely fast and are showing strong tendencies towards more advanced capabilities and structures, for example, using generative AI. These systems and tools are taking a whole new place at judge's desks: decision-making support. Even though the current literature underlines few investigations tackling AI's effects on judges' discretion [11], there's an indication of pressing consequences of this human-machine interaction in the field of Justice and fairness [3]. Working towards answering the main question, we pursued a set of research steps and sub-questions: firstly we observed processes of AI adoption in public services, we traced back historically the digital transformation of Brazil's Justice System and mapped out important issues regarding human-machine interaction. We also contextualize the current scenario of Brazilian Courts' technological developments, by means of official documents published by the National Council of Justice [14] and use it as a

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background for a discussion on the motives, incentives, strategies and overall readiness (or not) of the Judiciary for AI adoption. We raise the argument that the speed with which this has been made contrasts heavily with the lack of bureaucrats' AI training and knowledge regarding its risks when used for decision-making. Decision fairness and overall discretionary qualities inherent to a judge's role may be affected by the way these tools have been shaping and molding their work routines, especially considering the institutional pressure for productivity and case-solving goals. As for our methodology, we relied on documents analysis [6], composed of secondary sources and data on Brazil's Justice System as a whole and the current state of AI implementation in courts, all gathered by the National Council of Justice, as well as public manifestations of discourse regarding AI use and its advantages to bureaucrats' work by both courts specified above. A second stage relied on in-depth interviews [2], from which we gathered data on perceptions, practices and overall changes felt and reported by judges. Five judges accepted to participate and be interviewed, of which one was from the Federal Court and the other four from the State Court. All of them had a similar career time, around 15 years on the job, and differentiated themselves from their colleagues by means of a high personal interest in technology, which led them to understand it more profoundly. Our interview script covered mainly strategic and work-related changes brought about by the use of AI tools, and even provoked judges to imagine their work routines five years into the future. The questions resulted in 6 categories of thematic topics: 1) work overload and Judiciary innovation; 2) origins and continuity of digital transformations in the Judiciary; 3) tech uses in work routines and judges' training scope; 4) use dependance, proneness to error and emerging preoccupations on AI; 5) level of institutional incentive towards AI implementation; 6) judges' take on their works' future. As for our hypothesis, we tested two main questions: 1) Does the heavy workload and extensive backlog of cases predispose Brazilian judges to excessively trust decision-making support tools? 2) Does the trust in AI's positive results when used in low-risk contexts favours an anxiety to develop and implement more advanced AI in higher-risk contexts? Thematic analysis of those interviews revealed evidence that AI tools have indeed been frequently used by the judiciary on a day to day basis, including informal uses of Chatbots and Gen AI services, mainly to support decision-making. Results also point to a high probability of over-reliance on these tools' results by these judges, who expressed concerns regarding faults in AI but declared that they didn't have enough training to comprehend or deal with such tools and their errors. We gathered an overall ambiguous position towards how AI has affected judges' discretion, which raises important questions regarding how some tech-prepared courts may be seen as more "modern" and "productive" because of AI use and as such peer-pressure other unresourceful courts to also adopt these technologies despite not having enough tech capabilities. As a byproduct of this paper, we find that a fairly clear scenario can be described: the AI implementation race by public services in Brazil has been excessively rushed and lacks critical thinking regarding when, why and how to adopt these tools. This research field points to an agenda that needs to look further into political aspects of AI implementation, as well as social-cultural effects of AI as a distinctive trait to institutions and bureaucrats within their social circles. We believe that these results may impact the development of better governance models for public sectors taken by digitalization and modernization processes [13], and also help to better map transformations of work and of public institutions affected by AI as a whole.

Keywords: Artificial Intelligence, Justice System, Bureaucrats, Digitalization, Public Services

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References

- [1] Andersson, C., Hallin, A., & Ivory, C. 2022. *Unpacking the digitalisation of public services: Configuring work during automation in local government*. Government Information Quarterly, 39(1), 101662.
- [2] Barbot, J. 2015. *Conduzir uma entrevista de face a face*. In: Serge Paugam. A Pesquisa Sociológica, 102-123.
- [3] Binns, R. 2022. *Human judgment in algorithmic loops: individual justice and automated decision-making*. Regulation & Governance, v. 16, n. 1, 197-211.

- [4] Bovens, M., & Zouridis, S. 2002. *From street-level to system-level bureaucracies: How information and communication technology is transforming administrative discretion and constitutional control*. Public administration review, 62(2), 174-184.
- [5] Brandão, R. 2025. *Governança da Inteligência Artificial no setor público: capacidades técnico-administrativas vacilantes, falta de atenção aos municípios e polarização do debate - uma soma perigosa para a municipalização do reconhecimento facial no Brasil*. Thesis, PhD. Faculty of Philosophy, Languages and Social Sciences, University of São Paulo.
- [6] Cellard, A. 2012. *A análise documental*. In: Jean Poupart, A Pesquisa Qualitativa: Enfoques epistemológicos e metodológicos, 295-316.
- [7] De Boer, N., & Raaphorst, N. 2023. *Automation and discretion: explaining the effect of automation on how street-level bureaucrats enforce*. Public Management Review, 25(1), 42-62.
- [8] Kapoor, Sayash et al. 2024. *Promises and pitfalls of artificial intelligence for legal applications*. Journal of Cross-disciplinary Research in Computational Law, v.2, n.2.
- [9] Kerssens, N. & Dijck, J. van. 2021. *The platformization of primary education in The Netherlands*. Learning, Media and Technology, 46(3), 250–263.
- [10] Kolkman, D. et al 2024. *Justitia ex machina: The impact of an AI system on legal decision-making and discretionary authority*. Big Data & Society, v. 11, n. 2.
- [11] Janssen, M et al. 2020. *Will algorithms blind people? The effect of expainable AI and decision-makers' experience on AI-supported decision-making in government*. Social Science Computer Review, v. 40, n. 2, 478-493, SAGE Publications.
- [12] Maranhão, J et al. 2023. *Transparência sobre o emprego de inteligência artificial no Judiciário: um modelo de governança*. Suprema: revista de estudos constitucionais, Brasília, v. 3, n. 2, 145-187.
- [13] Mergel, I. et al. 2023. *Implementing AI in the public sector*. Public Management Review, [S.L.], p. 1-14.
- [14] National Council of Justice [Conselho Nacional de Justiça (CNJ)]. 2025. *Research on Artificial Intelligence in the Judiciary 2024: executive summary*. Available at: < https://www.cnj.jus.br/wp-content/uploads/2025/09/relatorio-ia-2024-resumo-executivo-set2025.pdf&ved=2ahUKEwiztejt-ZaUAxWJpJUCHaclKEAQFnoECBcQAQ&usg=AOvVaw3sH4g4BGmagGo_fUQo2e9X>
- [15] Nieto-Morales, F., Lotta, G., & Peeters, R. 2024. *Frontline work in weak institutions: implementing inequities. In Street-level bureaucracy in weak state institutions*. Policy Press, pp. 230-244.
- [16] Selten, Friso, Robeer, M., & Grimmelikhuijsen, S. 2023. *'Just like I thought': Street-level bureaucrats trust AI recommendations if they confirm their professional judgment*. Public Administration Review, 83(2), 263-278.
- [17] Ribeiro, M. M.; Segatto, C. I. 2025. *Inteligência artificial nas organizações públicas brasileiras: heterogeneidades e capacidades em tecnologia da informação*. Revista de Administração Pública, v. 59, n. 1, p. e2024-0066.